

Lab 8

Programing with Python Lab

BCACC393

Part A — Arrays in Python

1. Create and Access an Array

Write a Python program to create an array using the array module containing 10 integer values. Display all the elements of the array and access the first, last, and middle elements using indexing.

2. Array Operations

Create an integer array containing 10 numbers. Write a program to calculate and display the **sum, average, maximum, minimum, and length** of the array.

3. Insert and Delete Elements

Create an array containing the numbers from 1 to 10. Write a program to:

- Insert a new element at a specified position.
- Append an element at the end.
- Remove a specified element.
- Display the updated array.

4. Reverse and Sort an Array

Create an array of 10 randomly selected integers. Write a program to sort the array in **ascending and descending order** and display the reversed array.

5. Search and Count Elements

Create an array of integers and write a program to:

- Search for a particular element.
- Find the index of the element if it exists.
- Count the number of occurrences of the element.
- Display an appropriate message if the element is not found.

Part B — NumPy

6. Creating NumPy Arrays

Write a Python program using NumPy to create:

- A one-dimensional array.
- A two-dimensional array,

- An array containing zeros.
- An array containing ones.
- An array containing a range of numbers.

Display each array and its shape.

7. Array Indexing and Slicing

Create a NumPy array containing numbers from 1 to 20. Demonstrate:

- Positive indexing.
- Negative indexing.
- Extracting the first five elements.
- Extracting the last five elements.
- Extracting alternate elements.

8. NumPy Array Arithmetic

Create two NumPy arrays of equal size. Perform and display:

- Addition
- Subtraction
- Multiplication
- Division
- Modulus
- Exponentiation

9. Statistical Operations Using NumPy

Create a NumPy array containing marks of 10 students. Calculate:

- Mean
- Median
- Standard deviation
- Variance
- Maximum marks
- Minimum marks

10. Reshaping NumPy Arrays

Create a NumPy array containing numbers from 1 to 24. Reshape it into:

- A 2×12 matrix
- A 3×8 matrix
- A 4×6 matrix
- A 6×4 matrix

Display the shape of each resulting array.

11. Matrix Operations

Create two 3×3 NumPy matrices and perform:

- Matrix addition
- Matrix subtraction
- Element-wise multiplication
- Matrix multiplication
- Transpose of each matrix

12. Finding Elements Using NumPy Conditions

Create a NumPy array containing 20 integers. Write a program to:

- Find all even numbers.
- Find all odd numbers.
- Find numbers greater than 50.
- Find numbers between 20 and 60.

Use NumPy's Boolean indexing.

13. NumPy Mathematical Functions

Create a NumPy array containing angles in degrees. Use NumPy functions to calculate the corresponding:

- Sine values
- Cosine values
- Tangent values

Also demonstrate the use of `sqrt()`, `log()`, and `exp()` on a suitable array.

14. Combining and Splitting Arrays

Create two NumPy arrays and demonstrate:

- `concatenate()`
- `vstack()`
- `hstack()`
- `split()`

Display the resulting arrays after each operation.

15. Handling Missing and Invalid Values in NumPy

Create a NumPy array containing numeric values along with NaN values. Write a program to:

- Identify the NaN values.
- Count the missing values.
- Calculate the mean while ignoring NaN.
- Replace NaN values with the mean of the available values.

Part C — Pandas

16. Creating a Pandas Series

Create a Pandas Series containing the marks of 10 students. Display:

- The Series.
- Its index.
- Its values.
- The first five elements.
- The last three elements.
- The maximum and minimum marks.

17. Creating and Accessing a DataFrame

Create a Pandas DataFrame containing the following information for 10 students:

- Roll Number
- Name
- Department
- Marks
- Attendance

Perform operations to:

- Display the complete DataFrame.
- Display the first five records.
- Display the last five records.
- Display selected columns.
- Display a particular student's record.

18. DataFrame Filtering

Using a student DataFrame, write a program to display:

- Students who scored more than 75 marks.
- Students whose attendance is greater than 80%.
- Students who scored more than 60 and have attendance greater than 75%.
- Students belonging to a particular department.

19. Sorting and Ranking Data

Create a Pandas DataFrame containing employee information such as employee ID, name, department, and salary. Write a program to:

- Sort employees by salary in ascending order.
- Sort employees by salary in descending order.
- Find the highest-paid employee.
- Find the lowest-paid employee.
- Rank employees based on salary.

20. Handling Missing Data in Pandas

Create a DataFrame containing student information with some missing values in the marks, attendance, or department columns. Write a program to:

- Identify missing values.
- Count missing values in each column.
- Remove rows containing missing values.
- Replace missing numerical values with the column mean.

21. Grouping and Aggregation

Create a DataFrame containing employee details with **department, employee name, salary, and experience**. Use Pandas `groupby()` to calculate:

- Average salary for each department.
- Maximum salary for each department.
- Minimum salary for each department.
- Number of employees in each department.

Part D — Matplotlib

22. Line Plot

Using Matplotlib, create a line graph showing the **monthly sales of a company for 12 months**.

The graph should contain:

- Appropriate X-axis and Y-axis labels.
- A suitable title.
- Markers for data points.
- Grid lines.

23. Bar Chart

Create a bar chart showing the **marks obtained by five students in five different subjects**.

The chart should:

- Display student/subject names on the X-axis.
- Display marks on the Y-axis.
- Have an appropriate title.
- Display the values on top of the bars.

24. Multiple Plots and Subplots

Create a dataset containing the monthly temperature of a city for one year. Using Matplotlib, create:

- A line plot showing monthly temperature.
- A bar chart showing monthly temperature.
- A scatter plot showing the temperature distribution.

Arrange the three visualizations using subplots and provide suitable titles and axis labels.

25. Pandas + NumPy + Matplotlib Data Visualization

Create a Pandas DataFrame containing the following information for at least 10 students:

- Student Name
- Mathematics Marks
- Physics Marks
- Chemistry Marks
- Computer Science Marks

Use **Pandas** to calculate the average marks of each subject and use **NumPy** where appropriate for numerical calculations. Then use **Matplotlib** to create suitable visualizations such as:

- A bar chart comparing average marks in each subject.
- A line plot showing the marks of individual students.
- A histogram showing the distribution of total/average marks.

Add appropriate titles, axis labels, legends, and grid lines to the plots.